PTF ALL

Cabin Pressure Outflow Valve Does Not Operate (Manual Mode) - Fault Isolation

CORRECTIVE ACTION SUMMARY

Repair/Replace Bad Components for Cabin Pressure Outflow Valve [V48]
Replace Cabin Pressure Outflow Valve [V48]
Repair Wiring D10724-D10730
Repair Wiring D10724-Circuit Breaker C1269
Replace PRESSURIZATION CONTROL MANUAL Circuit Breaker [C1269]
Replace Cabin Pressure Control Module [P5-6]

A. Description

(1) This task is for this observed fault:
   (a) Manual mode control of cabin pressure outflow valve: does not operate.

(2) The aft outflow valve cannot be moved towards open or closed using the manual toggle switch and with the cabin pressure mode select switch in MAN.

B. Initial Evaluation

(1) For the cabin pressure outflow valve, V48, do this task:
   • Pressurization System Manual Mode Test, AMM TASK 21-31-00-710-801.
   (a) If the system does not operate as specified in the AMM task, then do the Fault Isolation Procedure below.
   (b) If the system operates as specified in the AMM task, then there was an intermittent fault.

C. Fault Isolation Procedure

(1) Do a detailed visual inspection of the cabin pressure outflow valve, V48 and its connector D10730P/D10730J for bent or pushed back pins/sockets. This is the task:
   • Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.
   (a) If you find a problem during the inspection, then do these steps:
      1) Repair or replace the bad components.
      2) Do the AMM test of the cabin pressure outflow valve that you did at the start of this task.
      3) If the system operates as specified in the AMM task, then you corrected the fault.
   (b) If you do not find a problem during the inspection, then continue.

(2) Do this check of the voltage at the cabin pressure outflow valve:
   (a) Disconnect connector D10730 from the cabin pressure outflow valve, V48.
   (b) Set the cabin pressure MANUAL toggle switch (flight deck) to OPEN.
   (c) Do a check for voltage between pin A and pin D of connector D10730 (WDM 21-31-11). The correct voltage is 48 VDC.
   (d) Set the cabin pressure MANUAL toggle switch (flight deck) to CLOSE.
   (e) Do a check for voltage between pin B and pin D of connector D10730 (WDM 21-31-11). The correct voltage is 48 VDC.
   (f) If the voltage is correct at both of the specified switch positions, then do these steps:
      1) Replace the cabin pressure outflow valve, V48. These are the tasks:
         • Aft Outflow Valve Assembly Removal, AMM TASK 21-31-03-000-801,
         • Aft Outflow Valve Assembly Installation, AMM TASK 21-31-03-400-801.
      2) Do the AMM test of the cabin pressure outflow valve that you did at the start of this task.
      3) If the system operates as specified in the AMM task, then you corrected the fault.
   (g) If the voltage is not correct at either of the specified switch positions, then continue.

(3) Do this check of the wiring:

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(a) Disconnect connector D10724 from the cabin pressure control module, P5-6.

(b) Do a wiring check between these pins of connector D10724 at the cabin pressure control module and these pins of connector D10730 at the cabin pressure outflow valve (WDM 21-31-11):

D10724		D10730
pin 25	.....	pin A
pin 24	.....	pin B

(c) Do a wiring check to make sure that pin 28 of D10724 has continuity to ground.

(d) Do a wiring check to make sure that pin D of D10730 has continuity to ground.

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D10724 to the cabin pressure control module, P5-6.

3) Re-connect connector D10730 to the cabin pressure outflow valve, V48.

4) Do the AMM test of the cabin pressure outflow valve that you did at the start of this task.

5) If the system operates as specified in the AMM task, then you corrected the fault.

(f) If you do not find a problem with the wiring, then continue.

(g) Re-connect connector D10730 to the cabin pressure outflow valve, V48.

(4) Do this check of the voltage at the cabin pressure control module:

(a) Do a check for voltage between pin 15 of connector D10724 and ground (WDM 21-31-11). The correct voltage is 28 VDC.

(b) If the correct voltage is not present, then do this check of the wiring:

1) Open the PRESSURIZATION CONTROL MANUAL circuit breaker, C1269 [P6-4, F6].

2) Do a wiring check between this pin of connector D10724 at the cabin pressure control module and this terminal of circuit breaker C1269 (WDM 21-31-11):

D10724		C1269
pin 15	.....	term =L

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D10724 to the cabin pressure control module, P5-6.

c) Close the PRESSURIZATION CONTROL MANUAL circuit breaker, C1269 [P6-4, F6].

d) Do the AMM test of the cabin pressure outflow valve that you did at the start of this task.

e) If the system operates as specified in the AMM task, then you corrected the fault.

4) If you do not find a problem with the wiring, then do these steps:

a) Re-connect connector D10724 to the cabin pressure control module, P5-6.

b) Close the PRESSURIZATION CONTROL MANUAL circuit breaker, C1269 [P6-4, F6].

c) Replace the PRESSURIZATION CONTROL MANUAL circuit breaker, C1269.

d) Do the AMM test of the cabin pressure outflow valve that you did at the start of this task.

e) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the correct voltage is present, then continue.

(d) Re-connect connector D10724 to the cabin pressure control module, P5-6.

(5) Replace the cabin pressure control module, P5-6. These are the tasks:

- Cabin Pressure Control Module Removal, AMM TASK 21-31-02-000-801,
- Cabin Pressure Control Module Installation, AMM TASK 21-31-02-400-801.

(a) Do the AMM test of the cabin pressure outflow valve that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

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